A

Matched native-SNN controls: hold Z or let its original dynamics continue

From the 8-s checkpoint

From the 9.4-s checkpoint

After the same external refill

Mean E-target Z

Native Z evolves

Frozen at the 8-s checkpoint

Frozen at the 9.4-s checkpoint

Held at Z = 1 after refill

E: native Z

E: fixed Z

I: native Z

I: fixed Z

Population rate (Hz; 5-ms bins)

Time (s)

B

Mean Z

M off · refill

ηM=0.2 · no refill

ηM=0.8 · no refill

Adaptation current (mV equiv.)

Depletion excess $q - (1 - \bar{Z})$

E rate (Hz) 0.5-s mean

Time (s)

C

η_M

τ_M (s)

Entry time / lower bound (s)